Year 10 Higher

REVISION PACK

Shape and Space 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Q	Working	Answer
the exterior angle of a triangle		
1.1	$x + 20 + 2x + 10 = 4x - 10$ $3x + 30 = 4x - 10$ $x = 40$ angle $ACD = 150^\circ$	**150°**
1.2	$2x + 15 + x + 20 = 4x - 10$ $3x + 35 = 4x - 10$ $x = 45$ angle $ACD = 170^\circ$	**170°**
1.3	$x + 15 + 3x - 5 = 5x - 20$ $4x + 10 = 5x - 20$ $x = 30$ angle $ACD = 130^\circ$	**130°**
1.4	$x + 30 + 2x + 5 = 4x - 5$ $3x + 35 = 4x - 5$ $x = 40$ angle $ACD = 155^\circ$	**155°**
1.5	$2x + 5 + x + 15 = 4x - 6$ $3x + 20 = 4x - 6$ $x = 26$ angle $ACD = 98^\circ$	**98°**
1.6	$x + 25 + x + 35 = 3x + 10$ $2x + 60 = 3x + 10$ $x = 50$ angle $ACD = 160^\circ$	**160°**
angles in a parallelogram		

1.7	$3x + 20 + 2x + 10 = 180$ (co-interior) $5x + 30 = 180$ $x = 30$ angles: 110° and 70°	**70°**
1.8	$3x - 15 + 2x + 5 = 180$ (co-interior) $5x - 10 = 180$ $x = 38$ angles: 99° and 81°	**81°**
1.9	$2x + 40 + 3x - 10 = 180$ (co-interior) $5x + 30 = 180$ $x = 30$ angles: 100° and 80°	**80°**
1.10	$4x + 10 + 2x + 20 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 110° and 70°	**70°**
1.11	$3x + 10 + 3x + 20 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 85° and 95°	**85°**
1.12	$2x + 25 + 4x + 5 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 75° and 105°	**75°**

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Q	Working	Answer
finding a shorter side		
2.1	$x^2 = 15.6^2 - 9.3^2 = 156.87$ $x = \sqrt{156.87} = 12.5248\ldots$	x **= 12.5 cm**
2.2	$x^2 = 11.8^2 - 6.4^2 = 98.28$ $x = \sqrt{98.28} = 9.91363\ldots$	x **= 9.91 cm**

2.3	$x^2 = 20.5^2 - 13.7^2 = 232.56$ $x = \sqrt{232.56} = 15.2499\ldots$	x **= 15.2 cm**
2.4	$x^2 = 9.4^2 - 5.1^2 = 62.35$ $x = \sqrt{62.35} = 7.8962\ldots$	x **= 7.9 cm**
2.5	$x^2 = 17.2^2 - 10.8^2 = 179.2$ $x = \sqrt{179.2} = 13.3866\ldots$	x **= 13.4 cm**
2.6	$x^2 = 13.5^2 - 8.9^2 = 103.04$ $x = \sqrt{103.04} = 10.1509\ldots$	x **= 10.2 cm**
the area of a trapezium		
2.7	$\frac{1}{2}(x + x + 3) \times 8 = 60$ $(2x + 3) \times 8 = 120$ $2x + 3 = 15$ $x = 6$	$x = 6$
2.8	$\frac{1}{2}(x + x + 4) \times 7 = 98$ $(2x + 4) \times 7 = 196$ $2x + 4 = 28$ $x = 12$	$x = 12$
2.9	$\frac{1}{2}(x + x + 6) \times 4 = 48$ $(2x + 6) \times 4 = 96$ $2x + 6 = 24$ $x = 9$	$x = 9$
2.10	$\frac{1}{2}(x + x + 6) \times 7 = 91$ $(2x + 6) \times 7 = 182$ $2x + 6 = 26$ $x = 10$	$x = 10$
2.11	$\frac{1}{2}(x + x + 4) \times 5 = 55$ $(2x + 4) \times 5 = 110$ $2x + 4 = 22$ $x = 9$	$x = 9$

2.12	$\frac{1}{2}(x+x+2)\times 8=120$ $(2x+2)\times 8=240$ $2x+2=30$ $x=14$	$x=14$
radius from the area		
2.13	$\pi r^2=120$ $r^2=120\div\pi=38.1972\ldots$ $r=\sqrt{38.1972\ldots}=6.18039\ldots$	**6.18 cm**
2.14	$\pi r^2=300$ $r^2=300\div\pi=95.493\ldots$ $r=\sqrt{95.493\ldots}=9.77205\ldots$	**9.77 cm**
2.15	$\pi r^2=85$ $r^2=85\div\pi=27.0563\ldots$ $r=\sqrt{27.0563\ldots}=5.20157\ldots$	**5.2 cm**
2.16	$\pi r^2=60$ $r^2=60\div\pi=19.0986\ldots$ $r=\sqrt{19.0986\ldots}=4.37019\ldots$	**4.37 cm**
2.17	$\pi r^2=250$ $r^2=250\div\pi=79.5775\ldots$ $r=\sqrt{79.5775\ldots}=8.92062\ldots$	**8.92 cm**
2.18	$\pi r^2=45$ $r^2=45\div\pi=14.3239\ldots$ $r=\sqrt{14.3239\ldots}=3.7847\ldots$	**3.78 cm**

Shape and Space 1 · Semicircles and compound shapes (Lesson 5.5)

Q	Working	Answer
semicircles and compound shapes		
3.1	(a) $14 + 10 + 14 + \frac{1}{2}\pi \times 10 = 38 + 15.70\ldots$ (b) $14 \times 10 + \frac{1}{2}\pi \times 5^2 = 140 + 39.26\ldots$	**(a) 53.7 cm** **(b) 179 cm²**
3.2	(a) $9 + 4 + 9 + \frac{1}{2}\pi \times 4 = 22 + 6.283\ldots$ (b) $9 \times 4 + \frac{1}{2}\pi \times 2^2 = 36 + 6.283\ldots$	**(a) 28.3 cm** **(b) 42.3 cm²**
3.3	(a) $20 + 12 + 20 + \frac{1}{2}\pi \times 12 = 52 + 18.84\ldots$ (b) $20 \times 12 + \frac{1}{2}\pi \times 6^2 = 240 + 56.54\ldots$	**(a) 70.8 cm** **(b) 297 cm²**
3.4	(a) $15 + \frac{1}{2}\pi \times 15 = 15 + 23.56\ldots$ (b) $\frac{1}{2}\pi \times 7.5^2 = 88.35\ldots$	**(a) 38.6 cm** **(b) 88.4 cm²**
3.5	(a) $16 + 7 + 16 + \frac{1}{2}\pi \times 7 = 39 + 10.99\ldots$ (b) $16 \times 7 + \frac{1}{2}\pi \times 3.5^2 = 112 + 19.24\ldots$	**(a) 50.0 cm** **(b) 131 cm²**
3.6	(a) $30 + 30 + \pi \times 10 = 60 + 31.41\ldots$ (b) $30 \times 10 + \pi \times 5^2 = 300 + 78.53\ldots$	**(a) 91.4 cm** **(b) 379 cm²**

Shape and Space 1 · Angles (Lesson 5.1)

Q	Working	Answer
angles in a quadrilateral		
3.7	$6x + 120 = 360$, $x = 40$ Angles: $120^\circ, 90^\circ, 60^\circ, 90^\circ$	**120°**
3.8	$6x = 360$, $x = 60$ Angles: $60^\circ, 90^\circ, 120^\circ, 90^\circ$	**120°**
3.9	$9x + 72 = 360$, $x = 32$ Angles: $128^\circ, 101^\circ, 79^\circ, 52^\circ$	**128°**
3.10	$10x + 110 = 360$, $x = 25$ Angles: $115^\circ, 90^\circ, 75^\circ, 80^\circ$	**115°**
3.11	$10x + 10 = 360$, $x = 35$ Angles: $85^\circ, 85^\circ, 95^\circ, 95^\circ$	**95°**
3.12	$5x + 140 = 360$, $x = 44$ Angles: $69^\circ, 83^\circ, 108^\circ, 100^\circ$	**108°**